



Estimated time needed: 15 minutes

In this lab, you will learn some commonly used DDL (Data Definition Language) statements of SQL. First you will learn the CREATE statement, which is used to create a new table in a database. Next, you will learn the ALTER statement which is used to add, delete, or modify columns in an existing table. Then, you will learn the TRUNCATE statement which is used to remove all rows from an existing table without deleting the table itself. Lastly, you will learn the DROP statement which is used to delete an existing table in a database.

How does the syntax of a CREATE statement look?

```
CREATE TABLE table_name (  
    column1 datatype,  
    column2 datatype,  
    column3 datatype,  
    ....  
);
```

How does the syntax of an ALTER statement look?

```
ALTER TABLE table_name  
ADD COLUMN column_name data_type column_constraint;  
ALTER TABLE table_name  
DROP COLUMN column_name;  
ALTER TABLE table_name  
ALTER COLUMN column_name SET DATA TYPE data_type;  
ALTER TABLE table_name  
RENAME COLUMN current_column_name TO new_column_name;
```

How does the syntax of a TRUNCATE statement look?

```
TRUNCATE TABLE table_name;
```

How does the syntax of a DROP statement look?

```
DROP TABLE table_name;
```

Software Used in this Lab

In this lab, you will use [IBM Db2 Database](#). Db2 is a Relational Database Management System (RDBMS) from IBM, designed to store, analyze and retrieve the data efficiently.

To complete this lab you will utilize a Db2 database service on IBM Cloud. If you did not already complete this lab task earlier in this module, you will not yet have access to Db2 on IBM Cloud, and you will need to follow this lab first:

- [Hands-on Lab : Sign up for IBM Cloud, Create Db2 service instance and Get started with the Db2 console](#)

Database Used in this Lab

The databases used in this lab are internal databases.

Objectives

After completing this lab, you will be able to:

- Create a new table in a database
- Add, delete, or modify columns in an existing table
- Remove all rows from an existing table without deleting the table itself
- Delete an existing table in a database

Instructions

When you approach the exercises in this lab, follow the instructions to run the queries on Db2:

- Go to the [Resource List](#) of IBM Cloud by logging in where you can find the Db2 service instance that you created in a previous lab under **Services** section. Click on the **Db2-xx service**. Next, open the Db2 Console by clicking on **Open Console** button. Click on the 3-bar menu icon in the top left corner and go to the **Run SQL** page. The Run SQL tool enables you to run SQL statements.
 - If needed, follow [Hands-on Lab : Sign up for IBM Cloud, Create Db2 service instance and Get started with the Db2 console](#)

Exercise 1: CREATE

In this exercise, you will use the CREATE statement to create two new tables using Db2.

1. You need to create two tables, **PETSALE** and **PET**. To create the two tables PETSAL and PET, copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**. In the **History** section below the editor box, you will be able to see if the query has been executed successfully or not.

```
CREATE TABLE PETSale (
  ID INTEGER NOT NULL,
  PET CHAR(20),
  SALEPRICE DECIMAL(6,2),
  PROFIT DECIMAL(6,2),
  SALEDATE DATE
);
```

```
CREATE TABLE PET (
  ID INTEGER NOT NULL,
  ANIMAL VARCHAR(20),
  QUANTITY INTEGER
);
```

The screenshot shows the IBM Db2 on Cloud web interface. On the left, a sidebar contains icons for various functions, with the 'SQL' icon highlighted by a red box. The main area is divided into two panes. The top pane, titled 'Data objects', shows a search bar and a list of objects, including 'DMT80331'. The bottom pane, titled 'History', shows a table of executed scripts. The 'Run all' button in the top right of the code editor is also highlighted with a red box.

SQL Code:

```
1 CREATE TABLE PETSale (
2   ID INTEGER NOT NULL,
3   PET CHAR(20),
4   SALEPRICE DECIMAL(6,2),
5   PROFIT DECIMAL(6,2),
6   SALEDATE DATE
7 );
8
9 CREATE TABLE PET (
10  ID INTEGER NOT NULL,
11  ANIMAL VARCHAR(20),
12  QUANTITY INTEGER
13 );
```

History Table:

Script	Date	Status	Runtime
Untitled - 1	Apr 21, 2023 4:03:57 PM	✓ 2	0.232 s
CREATE TABLE PETSale (ID INTEGER NOT NULL, PET CHAR(20), SALEPRIC...		✓	0.131 s
CREATE TABLE PET (ID INTEGER NOT NULL. ANIMAL VARCHAR(20). QUANTI...		✓	0.101 s

2. Now insert some records into the two newly created tables and show all the records of the two tables. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**.

```
INSERT INTO PETSale VALUES
(1, 'Cat', 450.09, 100.47, '2018-05-29'),
(2, 'Dog', 666.66, 150.76, '2018-06-01'),
```

```
(3, 'Parrot', 50.00, 8.9, '2018-06-04'),
(4, 'Hamster', 60.60, 12, '2018-06-11'),
(5, 'Goldfish', 48.48, 3.5, '2018-06-14');
```

```
INSERT INTO PET VALUES
```

```
(1, 'Cat', 3),
(2, 'Dog', 4),
(3, 'Hamster', 2);
```

```
SELECT * FROM PETSale;
```

```
SELECT * FROM PET;
```

IBM Db2 on Cloud

Data objects Saved objects

Find objects

SQL DMT80331

*Untitled ... x +

Syntax assistant

Run all

```
1 INSERT INTO PETSale VALUES
2 (1, 'Cat', 450.09, 100.47, '2018-05-29'),
3 (2, 'Dog', 666.66, 150.76, '2018-06-01'),
4 (3, 'Parrot', 50.00, 8.9, '2018-06-04'),
5 (4, 'Hamster', 60.60, 12, '2018-06-11'),
6 (5, 'Goldfish', 48.48, 3.5, '2018-06-14');
7
8 INSERT INTO PET VALUES
9 (1, 'Cat', 3),
10
```

History Results

Find history

Script	Date	Status	Runtime
Untitled - 1	Apr 21, 2023 4:08:05 PM	✓ 4	0.027 s
INSERT INTO PETSale VALUES (1, 'Cat', 450.09, 100.47, '2018-05-29'), (...		✓	0.010 s
INSERT INTO PET VALUES (1, 'Cat', 3), (2, 'Dog', 4), (3, 'Hamster', 2)		✓	0.007 s
SELECT * FROM PETSale		✓	0.005 s
SELECT * FROM PET		✓	0.005 s

You can click on the query in the History section to check its result:

Exercise 2: ALTER

Task A: ALTER using ADD COLUMN

- ```
ALTER TABLE PETSale
ADD COLUMN QUANTITY INTEGER;
SELECT * FROM PETSale;
```

The screenshot shows the IBM Db2 on Cloud interface. On the left, the 'Data objects' pane shows a database named 'DMT80331'. The main area is a SQL editor with a file named '\*Untitled ...'. The editor contains the following SQL code:

```
1 ALTER TABLE PETSale
2 ADD COLUMN QUANTITY INTEGER;
3
4 SELECT * FROM PETSale;
```

Below the editor, the 'Results' pane shows 'Result set 1' with a table of 5 rows. The table has columns: ID, PET, SALEPRICE, PROFIT, SALEDATE, and QUANTITY. The data is as follows:

| ID | PET      | SALEPRICE | PROFIT | SALEDATE   | QUANTITY |
|----|----------|-----------|--------|------------|----------|
| 1  | Cat      | 450.09    | 100.47 | 2018-05-29 |          |
| 2  | Dog      | 666.66    | 150.76 | 2018-06-01 |          |
| 3  | Parrot   | 50.00     | 8.90   | 2018-06-04 |          |
| 4  | Hamster  | 60.60     | 12.00  | 2018-06-11 |          |
| 5  | Goldfish | 48.48     | 3.50   | 2018-06-14 |          |

The 'Total:5' is displayed at the top right of the results table.

2. Now update the newly added **QUANTITY** column of the **PETSale** table with some values and show all the records of the table. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**. After the query has executed successfully, click on it to check the result set.

```
UPDATE PETSale SET QUANTITY = 9 WHERE ID = 1;
UPDATE PETSale SET QUANTITY = 3 WHERE ID = 2;
UPDATE PETSale SET QUANTITY = 2 WHERE ID = 3;
UPDATE PETSale SET QUANTITY = 6 WHERE ID = 4;
UPDATE PETSale SET QUANTITY = 24 WHERE ID = 5;
SELECT * FROM PETSale;
```



The screenshot shows the IBM Db2 on Cloud console interface. On the left, there is a sidebar with icons for navigation. The main area is divided into two sections: 'Data objects' and 'Saved objects'. The 'Data objects' section shows a search bar and a list of objects, including 'DMT80331'. The 'Saved objects' section is currently empty. The main area also displays a SQL editor with a query: `ALTER TABLE PETSale  
DROP COLUMN PROFIT;  
  
SELECT * FROM PETSale;`. The query is highlighted in blue. Below the editor, there are tabs for 'History' and 'Results'. The 'Results' tab is active, showing a table with 5 rows and 5 columns: ID, PET, SALEPRICE, SALEDATE, and QUANTITY. The table contains data for 5 different pets: Cat, Dog, Parrot, Hamster, and Goldfish. The 'Run all' button is visible in the top right corner of the console.

| ID | PET      | SALEPRICE | SALEDATE   | QUANTITY |
|----|----------|-----------|------------|----------|
| 1  | Cat      | 450.09    | 2018-05-29 | 9        |
| 2  | Dog      | 666.66    | 2018-06-01 | 3        |
| 3  | Parrot   | 50.00     | 2018-06-04 | 2        |
| 4  | Hamster  | 60.60     | 2018-06-11 | 6        |
| 5  | Goldfish | 48.48     | 2018-06-14 | 24       |

### Task C: ALTER using ALTER COLUMN

1. Change the data type to **VARCHAR(20)** type of the column **PET** of the table **PETSale** and show the altered table. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**.

```
ALTER TABLE PETSale
ALTER COLUMN PET SET DATA TYPE VARCHAR(20);
SELECT * FROM PETSale;
```

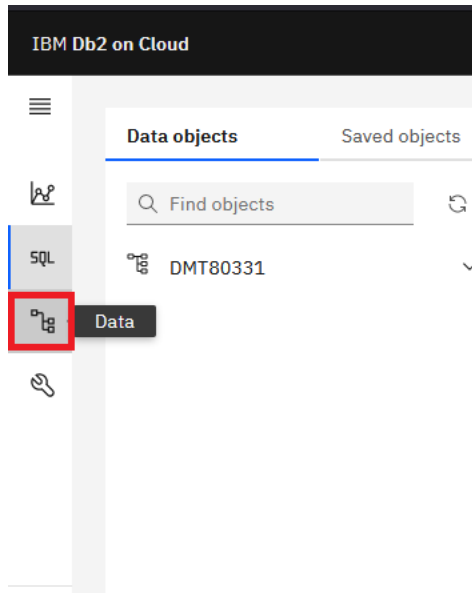
The screenshot displays the IBM Db2 on Cloud web interface. On the left, a sidebar contains navigation icons, with 'SQL' selected. The main area is divided into a 'Data objects' pane on the left and a central workspace. The workspace shows a SQL editor with the following code:

```
1 ALTER TABLE PETSale
2 ALTER COLUMN PET SET DATA TYPE VARCHAR(20);
3
4 SELECT * FROM PETSale;
```

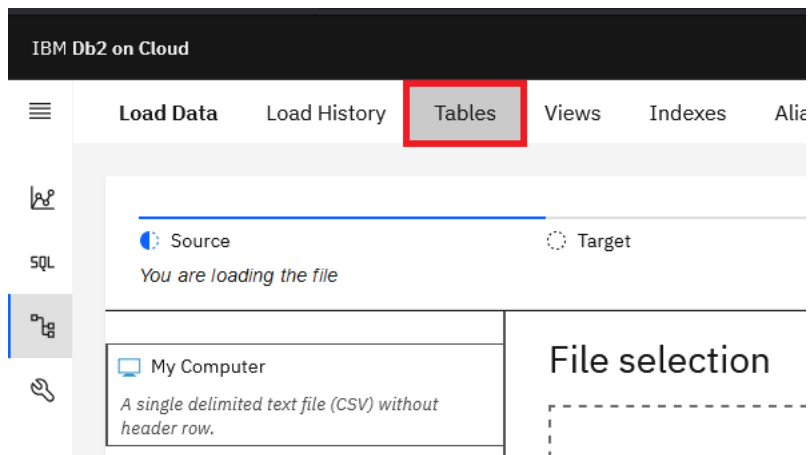
Below the editor, the 'Results' tab is active, showing 'Result set 1' with a table of 5 rows. The table has columns: ID, PET, SALEPRICE, SALEDATE, and QUANTITY. The data is as follows:

| ID | PET      | SALEPRICE | SALEDATE   | QUANTITY |
|----|----------|-----------|------------|----------|
| 1  | Cat      | 450.09    | 2018-05-29 | 9        |
| 2  | Dog      | 666.66    | 2018-06-01 | 3        |
| 3  | Parrot   | 50.00     | 2018-06-04 | 2        |
| 4  | Hamster  | 60.60     | 2018-06-11 | 6        |
| 5  | Goldfish | 48.48     | 2018-06-14 | 24       |

2. Now verify if the data type of the column **PET** of the table **PETSale** changed to **VARCHAR(20)** type or not. Click on the Data Section in the left menu bar.



Then click on Tables:



Find your schema and choose the table **PETSALE**

**IBM Db2 on Cloud**

Load DataLoad History**Tables**ViewsIndexesAliasesMQTsSequencesApplication objects

Find schemas or tables

Refresh

**Schemas**

| <input checked="" type="checkbox"/> | Name     | Definer type | Tables ▲ |
|-------------------------------------|----------|--------------|----------|
| <input checked="" type="checkbox"/> | DMT80331 | User         | 3        |

Total: 1, selected: 1

**Tables**

New table +

| <input type="checkbox"/> | Name ▼       | Schema   | Properties |
|--------------------------|--------------|----------|------------|
| <input type="checkbox"/> | BILLING_TEST | DMT80331 | ...        |
| <input type="checkbox"/> | PET          | DMT80331 | ...        |
| <input type="checkbox"/> | PETSALE      | DMT80331 | ...        |

Total: 3, selected: 0

You will see that the datatype of the column **PET** has changed to **VARCHAR(20)**

IBM Db2 on Cloud

Load DataLoad HistoryTablesViewsIndexesAliasesMQTsSequencesApplication objects

Find schemas or tables

Refresh

SQL

Schemas

Tables

Name

Schema

Properties

BILLING\_TEST

DMT80331

...

PET

DMT80331

...

PETSALE

DMT80331

...

Total: 3, selected: 0

New table

Table definition

PETSALE

Approximate 5 rows (32.0 KB)  
Updated on 2023-04-21 10:45:45

| Name      | Data type | Nullable | Length | Scale |  |
|-----------|-----------|----------|--------|-------|--|
| ID        | INTEGER   | N        |        | 0     |  |
| PET       | VARCHAR   | Y        | 20     | 0     |  |
| SALEPRICE | DECIMAL   | Y        | 6      | 2     |  |
| SALEDATE  | DATE      | Y        | 4      | 0     |  |
| QUANTITY  | INTEGER   | Y        |        | 0     |  |

View data

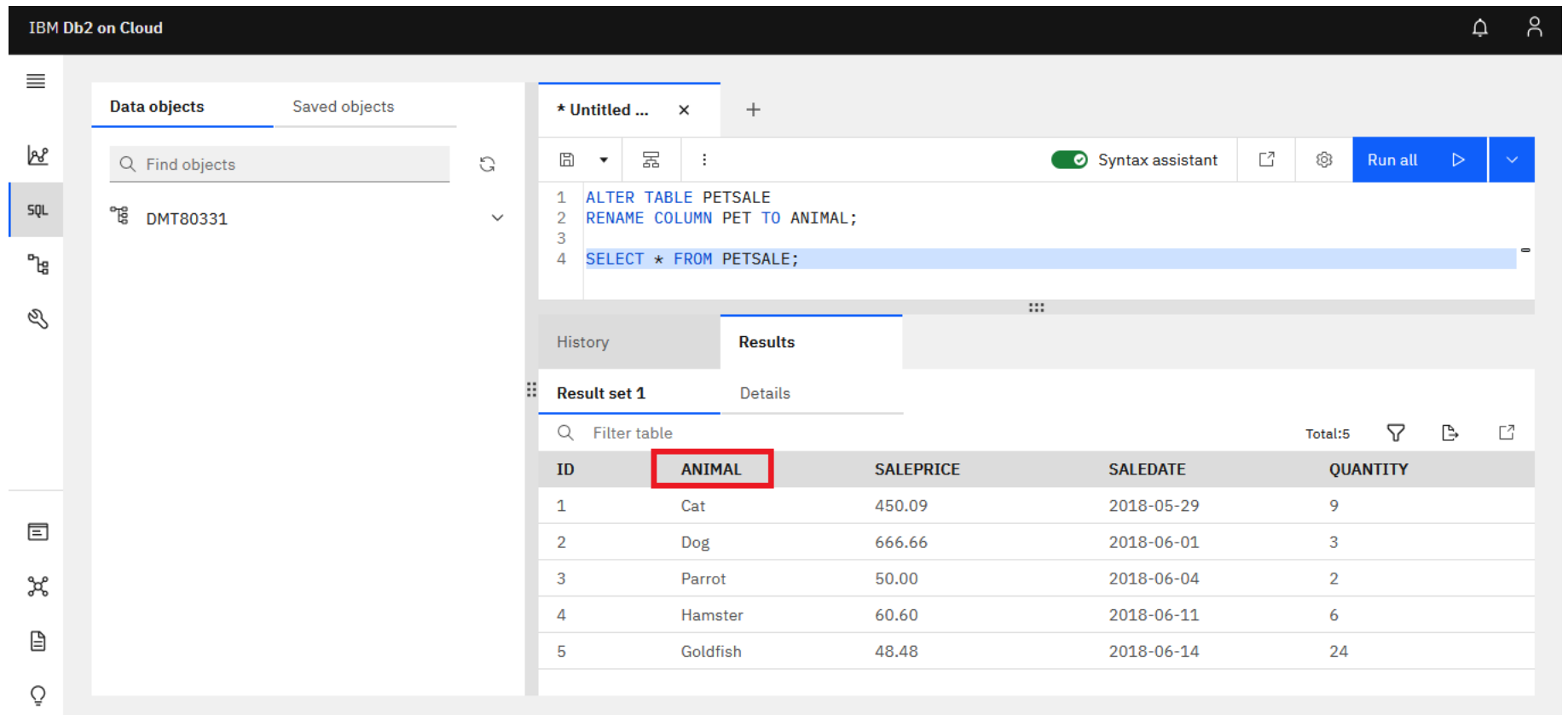
### Task D: ALTER using RENAME COLUMN

1. In the **PETSALE** table, rename the column **PET** to **ANIMAL** and show the altered table. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**.

```
ALTER TABLE PETSALE
RENAME COLUMN PET TO ANIMAL;
SELECT * FROM PETSALE;
```

about:blank

12/15



IBM Db2 on Cloud

Data objects Saved objects

Find objects

SQL

DMT80331

\*Untitled ... x +

Syntax assistant

Run all

```
1 ALTER TABLE PETSale;
2 RENAME COLUMN PET TO ANIMAL;
3
4 SELECT * FROM PETSale;
```

History Results

Result set 1 Details

Filter table

Total:5

| ID | ANIMAL   | SALEPRICE | SALEDATE   | QUANTITY |
|----|----------|-----------|------------|----------|
| 1  | Cat      | 450.09    | 2018-05-29 | 9        |
| 2  | Dog      | 666.66    | 2018-06-01 | 3        |
| 3  | Parrot   | 50.00     | 2018-06-04 | 2        |
| 4  | Hamster  | 60.60     | 2018-06-11 | 6        |
| 5  | Goldfish | 48.48     | 2018-06-14 | 24       |

In this exercise, you will use the TRUNCATE statement to remove all rows from an existing table created in exercise 1 without deleting the table itself.

1. Remove all rows from the **PET** table and show the empty table. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**. You will see **no data in the Result section**.

```
TRUNCATE TABLE PET IMMEDIATE;
SELECT * FROM PET;
```

The screenshot shows the IBM Db2 on Cloud console interface. On the left is a sidebar with navigation icons. The main area is divided into two panes. The left pane, titled 'Data objects', shows a search bar and a list of objects including 'DMT80331'. The right pane is the SQL editor, titled '\*Untitled ...', containing two SQL statements: 'TRUNCATE TABLE PET IMMEDIATE;' and 'SELECT \* FROM PET;'. Below the editor is a 'Results' pane showing 'Result set 1' with a table structure: ID, ANIMAL, QUANTITY. The table is currently empty, displaying a message: 'You don't have any data currently'.

In this exercise, you will use the DROP statement to delete an existing table created in exercise 1.

1. Delete the **PET** table and verify if the table still exists or not (SELECT statement won't work if a table doesn't exist). Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**. You will see that the **select statement fails**.

```
DROP TABLE PET;
SELECT * FROM PET;
```

IBM Db2 on Cloud

Data objects Saved objects

Find objects

SQL DMT80331

\*Untitled ... x +

Syntax assistant

Run all

```
1 DROP TABLE PET;
2
3 SELECT * FROM PET;
```

History Results

Find history

| Script            | Date                    | Status  | Runtime |
|-------------------|-------------------------|---------|---------|
| Untitled - 1      | Apr 21, 2023 4:28:00 PM | ✓ 1 ✗ 1 | 0.071 s |
| DROP TABLE PET    |                         | ✓       | 0.051 s |
| SELECT * FROM PET |                         | ✗       | 0.020 s |

**Congratulations! You have completed this Lab. You are ready for the next topic.**

**Author(s)**

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# Skills Network